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First Name(s):																

Department of Mathematics and Applied Mathematics

MAM2000W: 2LA (Linear Algebra)

July exam **6 July 2021**

Time : 2 hours **Maximum marks: 60**

Full marks: 60

- This question paper consists of 8 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators also allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
7	7	4	14	14	8	6	**60**

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Section A

In this section, give your final answer only. Do not show your working.

Question 1. [7 points] The matrices B and U are given below. The matrix A is the matrix B with its last column deleted.

$$B = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 & a \\ -4 & -1 & 0 & -6 & 2 & b \\ 0 & 3 & -6 & 3 & 1 & c \\ 2 & -1 & 3 & 6 & 0 & d \end{pmatrix} \qquad U = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 & a \\ 0 & -1 & 2 & 0 & 0 & b+2a \\ 0 & 0 & 0 & 3 & 1 & c+3b+6a \\ 0 & 0 & 0 & 0 & 0 & d-9a-4b-c \end{pmatrix}$$

The matrix U is a row-echelon form of B (you need not check this) and $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 3 \\ k \end{pmatrix}$, where k is a constant.

(a) For which value(s) of k will the system of linear equations $A\mathbf{x} = \mathbf{b}$ have (i) no solutions (ii) a unique solution (iii) infinitely many solutions?

(i) no solution: $k \neq -2$ (ii) unique solution: not possible (iii) infinitely many solutions: $k = -2$

(b) Give the rank and nullity of A.

$\text{Rank}(A) = 3, \quad \text{Nullity}(A) = 2.$

(c) Give a basis for (i) $NS(A)$ (ii) $CS(A)$.

A basis for $NS(A)$ is $\left\{ \begin{pmatrix} -1/2 \\ 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1/3 \\ 1 \end{pmatrix} \right\}$

A basis for $CS(A)$ is $\left\{ \begin{pmatrix} 2 \\ -4 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ -6 \\ 3 \\ 6 \end{pmatrix} \right\}$

Question 2. [7 points] You are given that $T : P_2(x) \to M_{2\times 2}(\mathbb{R})$ is a linear transformation and that

$$T(1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T(x) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad T(x^2) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

(a) Find $T(ax^2 + bx + c)$.

$$T(ax^2 + bx + c) = \begin{pmatrix} 2a - c & b \\ -b & 2a - c \end{pmatrix}$$

(b) Give a basis for the range of T.

A basis for $\text{ran}(T)$ is $\left\{\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\right\}$.

(c) Give the kernel of T and a basis for it.

$\ker(T) = \{ax^2 + 2a : a \in \mathbb{R}\}$. A basis is $\{x^2 + 2\}$.

(d) Find the matrix A representing T with respect to the ordered bases

$$B = \{1, x, x^2\} \text{ and } C = \left\{\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}, \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}, \begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}, \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}\right\}$$

for $P_2(x)$ and $M_{2\times 2}(\mathbb{R})$ respectively.

$A = \begin{pmatrix}-1 & 0 & 2\\ 0 & 1 & 0\\ 0 & -1 & 0\\ -1 & 0 & 2\end{pmatrix}$.

Question 3. **[4 points]** For every $p(x), q(x) \in P_2(x)$, put $\langle p(x), q(x)\rangle = \int_0^1 p(x)q(x)\,dx$. You may assume that this defines an inner product on $P_2(x)$. Let $r(x) = x^2$

(a) Let $p(x) = ax^2 + bx + c$. Find $\langle r(x), p(x)\rangle$ in terms of a, b and c.

$\frac{a}{5} + \frac{b}{4} + \frac{c}{3}$

(b) Find $||r(x)||$.

$\sqrt{\frac{1}{5}}$

(c) The subset A of P_2 is defined by $A = \{ax^2+bx+c \in P_2(x) : (ax^2+bx+c) \perp r(x)\}$. An incomplete description of A is given below; there should be an equation in the place of the dots ....

$$A = \{ax^2 + bx + c \in P_2(x) : \ \ldots\}.$$

Write down this equation.

$A = \{ax^2 + bx + c \in P_2(x) : \frac{a}{5} + \frac{b}{4} + \frac{c}{3} = 0\}$

Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. **[14 points]** Let $S = \{A \in M_{2\times 2}(\mathbb{R}) : \det(A - A^T) = 0\}$ and

$$B = \left\{\begin{pmatrix}1 & -1\\ 1 & 0\end{pmatrix}, \begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}, \begin{pmatrix}0 & 1\\ -1 & 1\end{pmatrix}\right\}.$$

Prove or disprove each of the following statements:

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(**a**) S is closed under scalar multiplication.

True.
Let $A \in S$ and $\lambda \in \mathbb{R}$.
Then $\det(A - A^T) = 0$.
Hence $\det(\lambda A - (\lambda A)^T) = \det(\lambda A - \lambda A^T) = \det(\lambda(A - A^T)) = \lambda^2 \det(A - A^T) = \lambda^2 \cdot 0 = 0$.
It follows that $\lambda A \in S$.

(**b**) S is a subspace of $M_{2\times 2}(\mathbb{R})$.

True.
S is nonempty and closed under addition and scalar multiplication.

(**c**) $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \in \operatorname{span}(B)$.

False. The equation

$$\lambda_1 \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$

leads to a system of equations with augmented matrix
$\begin{pmatrix} 1 & -1 & 0 & -1 \\ -1 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$ which reduces to $\begin{pmatrix} 1 & -1 & 0 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.
This shows that the system is inconsistent, and hence $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \notin \operatorname{span}(B)$.

Alternatively: We can see that any matrix in the span of B must be of the form $\begin{pmatrix} a & b \\ -b & c \end{pmatrix}$.

(**d**) B is linearly independent.

True. If the right-hand side of the equation in the answer to (c) is replaced by the zero matrix, the reduced matrix of the resulting homogeneous system shows that the system has a unique solution $\lambda_1 = \lambda_2 = \lambda_3 = 0$.

(**e**) B is a basis for $M_{2\times 2}(\mathbb{R})$.

False. $M_{2\times 2}(\mathbb{R})$ has dimension 4, and hence every basis for it has 4 elements, while B has only 3.

(**f**) B is a basis for span (B).

True. B is linearly independent and spans $\operatorname{span}(B)$.

(**g**) $B \cup \left\{ \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ is a basis for $M_{2\times 2}(\mathbb{R})$.

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True. Since any matrix in the span of B must be of the form $\begin{pmatrix} a & b \\ -b & c \end{pmatrix}$, we have that $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$ is not in the span of B. Also note that B is linearly independent. Hence by Theorem 2.1.19, it follows that $B \cup \left\{ \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ is also linearly independent. Hence $B \cup \left\{ \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ is a set with 4 linearly independent vectors in $M_{2\times 2}(\mathbb{R})$, which is of dimension 4. Hence $B \cup \left\{ \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ is a basis for $M_{2\times 2}(\mathbb{R})$.

Question 5. [14 points] Let $A = \begin{pmatrix} 4 & 0 & 2 \\ 0 & -5 & 0 \\ -4 & 0 & -2 \end{pmatrix}$.

(a) Find all the eigenvalues of A.

We get the characteristic equation

$$(-5-\lambda)\left[(4-\lambda)(-2-\lambda)+8\right] = 0,$$

which simplifies to

$$(-5-\lambda)\lambda(\lambda-2) = 0.$$

Hence the eigenvalues are -5, 0, and 2.

(b) Give reasons for your answers to the following two questions, using only your answer to (a).

(i) Can A be written as a product of elementary matrices?

No. Since 0 is an eigenvalue, A is singular and can therefore not be written as a product of elementary matrices.

(ii) Can A be diagonalized?

Yes. A has 3 distinct eigenvectors and hence 3 linearly independent eigenvectors.

(c) For each eigenvalue, find a basis for the corresponding eigenspace.

A basis for W_{-5} is $\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$.

A basis for W_0 is $\left\{ \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \right\}$.

A basis for W_2 is $\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$.

(d) Find, if possible, matrices P and D such that D is diagonal and $P^{-1}AP = D$.

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We can take

$$P = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & -2 & -1 \end{pmatrix}.$$

Then

$$P^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & -1 \\ 2 & 0 & 1 \end{pmatrix}$$

and

$$D = P^{-1}AP = \begin{pmatrix} -5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

(**e**) Find an elementary matrix A and an upper triangular matrix T such that $A = ET$.

$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$ and $T = \begin{pmatrix} 4 & 0 & 2 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Question 6. [8 points] Read the following proof and then answer the questions following it.

Let A be a square matrix and $\lambda_1, \lambda_2, \lambda_3$ be distinct eigenvalues of A with corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

It can be shown that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent. (1)
Suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent.
Then there are scalars μ_1, μ_2, μ_3, not all zero, such that

$$\mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \mu_3\mathbf{v}_3 = \mathbf{0} \tag{2}$$

Then $A(\mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \mu_3\mathbf{v}_3) = A\mathbf{0}$ (3)
and hence

$$\mu_1\lambda_1\mathbf{v}_1 + \mu_2\lambda_2\mathbf{v}_2 + \mu_3\lambda_3\mathbf{v}_3 = \mathbf{0} \tag{4}$$

Multiply (2) by λ_3 and subtract (4) from it to obtain

$$\mu_1(\lambda_3 - \lambda_1)\mathbf{v}_1 + \mu_2(\lambda_3 - \lambda_2)\mathbf{v}_2 = \mathbf{0} \tag{5}$$

From this it follows that $\mu_1(\lambda_3 - \lambda_1) = \mu_2(\lambda_3 - \lambda_2) = 0$, (6)
and hence $\mu_1 = \mu_2 = 0$. (7)
Substituting (7) in (2) yields $\mu_3\mathbf{v}_3 = 0$, and hence $\mu_3 = 0$. (8)
But this is a contradiction, (9)
and hence $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ must be linearly independent.

(**a**) Explain how the equation (4) follows from equation (3).

For $i = 1, 2, 3$, we have $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ since $\mathbf{v}_i$ is an eigenvector of A corresponding to the eigenvalue λ_i. Then

$$\mathbf{0} = A\mathbf{0} = A(\mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \mu_3\mathbf{v}_3) = \mu_1A\mathbf{v}_1 + \mu_2A\mathbf{v}_2 + \mu_3A\mathbf{v}_3 = \mu_1\lambda_1\mathbf{v}_1 + \mu_2\lambda_2\mathbf{v}_2 + \mu_3\lambda_3\mathbf{v}_3.$$

(**b**) What is used to deduce (6) from (5)?

The linear independence of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

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(**c**) Where in the proof is the fact that the eigenvalues are distinct used, and how is it used?

It is used in deducing (7) from (6). Since $\lambda_1 \neq \lambda_3$, $\mu_1(\lambda_1 - \lambda_3) = 0$ implies $\mu_1 = 0$. Similarly, $\lambda_2 \neq \lambda_3$ implies $\mu_2 = 0$.

(**d**) Why can we say at (8) that if $\mu_3 \mathbf{v}_3 = 0$, then $\mu_3 = 0$?

This follows from the fact that since $\mathbf{v}_3$ is an eigenvector, $\mathbf{v}_3 \neq \mathbf{0}$.

(**e**) What is the contradiction referred to at (9)?

We assumed in (2) that not all the scalars μ_1, μ_2, μ_3 are 0. But we have now shown, in (7) and (8), that they are all 0.

(**f**) Prove the statement at (1). [Hint: Use a proof by contradiction.]

Suppose that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent. Note that $\mathbf{v}_1 \neq \mathbf{0}$ and $\mathbf{v}_2 \neq \mathbf{0}$ since $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors.
Hence there is a non-zero scalar α such that $\mathbf{v}_1 = \alpha \mathbf{v}_2$.
Since $\mathbf{v}_1$ is eigenvector corresponding to λ_1, it holds that

$$A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1.$$

On the other hand, since $\mathbf{v}_2$ is eigenvector corresponding to λ_2, and $\mathbf{v}_1 = \alpha \mathbf{v}_2$, we get

$$A\mathbf{v}_1 = A(\alpha \mathbf{v}_2) = \alpha A \mathbf{v}_2 = \alpha \lambda_2 \mathbf{v}_2 = \alpha \lambda_2 \frac{1}{\alpha} \mathbf{v}_1 = \lambda_2 \mathbf{v}_1.$$

It follows that

$$\lambda_1 \mathbf{v}_1 = \lambda_2 \mathbf{v}_1,$$

and hence

$$(\lambda_1 - \lambda_2)\mathbf{v}_1 = \mathbf{0}.$$

But this is impossible, since $\mathbf{v}_1 \neq \mathbf{0}$ and $\lambda_1 \neq \lambda_2$.

Question 7. [6 points] A square matrix B is called *nilpotent* if there is a positive integer k such that B^k is the zero matrix.

(**a**) Give an example of a non-zero 2×2 nilpotent matrix.

$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$ are possible correct answers.

(**b**) Prove that if B is nilpotent, then 0 is an eigenvalue of B, and in fact the only eigenvalue.

True.
If B is a zero matrix then the statement is easily seen to be true.
If B is a non-zero nilpotent matrix then there is an integer $k > 1$ such that B^k is the zero matrix, but B^{k-1} is non-zero. Hence there is a vector $\mathbf{v}$ such that $B^{k-1}\mathbf{v} \neq \mathbf{0}$. It holds that

$$B(B^{k-1}\mathbf{v}) = (BB^{k-1})\mathbf{v} = B^k\mathbf{v} = \mathbf{0}\mathbf{v} = \mathbf{0} = 0(B^{k-1}\mathbf{v}).$$

Hence $B^{k-1}\mathbf{v}$ is eigenvector of B corresponding to eigenvalue 0.
Let now λ be any eigenvalue of B. Then there is a vector $\mathbf{u} \neq \mathbf{0}$ such that $B\mathbf{u} = \lambda\mathbf{u}$. But then

$$B^k\mathbf{u} = \lambda^k\mathbf{u}.$$

On the other hand,

$$B^k\mathbf{u} = \mathbf{0}\mathbf{u} = \mathbf{0}.$$

Hence $\lambda^k\mathbf{u} = \mathbf{0}$. Since $\mathbf{u} \neq \mathbf{0}$, this means that $\lambda^k = 0$, and hence $\lambda = 0$.

(**c**) Prove that if a 2×2 matrix has characteristic equation $\lambda^2 = 0$, then it is nilpotent.

Let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be any 2×2 matrix. Then the characteristic equation for B is

$$\lambda^2 - (a+d)\lambda + (ad - bc) = 0.$$

If the characteristic equation of B is

$$\lambda^2 = 0$$

then it most hold that $a + d = 0$ and $ad - bc = 0$.
Let's determine B^2. We have

$$B^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix}$$

Since $d = -a$, we get

$$\begin{aligned} B^2 &= \begin{pmatrix} a^2 + bc & ab + b(-a) \\ ac + c(-a) & bc + (-a)^2 \end{pmatrix} \\ &= \begin{pmatrix} a^2 + bc & 0 \\ 0 & bc + a^2 \end{pmatrix} \end{aligned}$$

Finally, note that $a + d = 0$ and $ad - bc = 0$ together imply that $a^2 + bc = 0$. It follows that

$$B^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

and hence B is nilpotent.

(**d**) Prove or disprove: A diagonalizable square matrix with a non-zero eigenvalue cannot be nilpotent.

True.
Follows directly from the fact that (b) is true.